

ENAE 464

Flow Over Smooth Cylinder

The objective of this experiment is to study the flow over a smooth cylinder by measuring surface pressure.

Experiments

One body is provided for this experiment: a circular cylinder with diameter of 1.27 cm. The cylinder has a single pressure tap on its surface. The cylinder can be mounted so that the pressure tap is at any desired angle to the flow.

Procedure:

1. At one flow speed, measure the pressure over the surface of the cylinder.
2. Use the pressure reading at the leading edge of the cylinder and Bernoulli's equation to determine the upstream flow speed.

Report:

Your report should conform to the writing guidelines discussed in the handout, and it should include the following items:

1. Summarize the inviscid flow theory for flow over a 2D cylinder. Be sure to identify all key assumptions and concepts. Derive the coefficient of pressure for the cylinder surface. You may follow the conformal mapping approach outlined in lecture, or summarize the approach given in Anderson's Fundamentals of Flight (Chapter 3). You only need to show key steps. Routine algebraic manipulations can be omitted and/or summarized in words.
2. A tabulation of operating conditions for the experiment, including ambient room temperature, and pressure.
3. Calculation of ambient air density from room temperature and pressure.
4. A table of the measured relative pressure, $P - P_\infty$, and the corresponding pressure coefficient, $C_p = 2(P - P_\infty)/(\rho U_\infty^2)$, versus angular position (P versus θ). The quantity $P - P_\infty$ is the pressure reading on the surface of the cylinder relative to the freestream pressure upstream of the cylinder. This quantity is directly given by the differential pressure transducer. U_∞ is the mean freestream velocity, which is obtained using the pitot-static tube (see next item).
5. Calculation of the mean wind tunnel velocity used for the experiment based on the pressure reading from the pitot-static tube in the wind tunnel.
6. A plot of C_p vs θ from the experiment, plotted simultaneous (on the same graph) with the pressure coefficient that is given by inviscid theory. Characterize any observed discrepancies, and explain the cause of these differences. How does your plot compare with published results?
7. Compute coefficient of drag using measured surface pressure. Does this agree with theoretical value? If not, explain why.

